MODULE 07 · DOMAIN 1 - MONITORING, LOGGING, ANALYSIS, REMEDIATION, AND PERFORMANCE OPTIMIZATION

Systems Manager for Operations

One prerequisite, then a dozen tools. Get the prerequisite wrong and none of them work.

DOMAIN 1 of 5 EXAM WEIGHT 22% MODULE 7 of 35

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Systems Manager is not one service. It is a dozen tools sharing **one prerequisite**: the node must be manageable.

What makes a node manageable

01

Agent

SSM Agent installed and running. Pre-installed on most modern AMIs.

02

Permissions

An instance profile, or account-level Default Host Management.

03

Network path

Reach ssm, ssmmessages and ec2messages endpoints.

04

Managed node

It appears in Fleet Manager, and every tool works.

ON THE EXAM

Agent, permissions, network. Three things. Every "it is not showing up" question is one of these three.

The modern way to grant permissions

- **Default Host Management Configuration** is an account and Region setting. Every eligible instance becomes managed, with no instance profile.
- AWS now **recommends this account-level approach** over attaching a profile to each instance.
- **It requires IMDSv2.** Instances still on IMDSv1 are not picked up, and there is no warning.
- It also needs **SSM Agent 3.2.582.0 or later**. Older agents are silently ignored.

> ON THE EXAM
>
> The classic answer is AmazonSSMManagedInstanceCore on an instance profile. That still works. DHMC is the account-wide version, and it has two hard prerequisites.

"My instance is not a managed node"

- **1. Is the agent running?** Check the service. Old AMIs may not have it at all.
- **2. Are there permissions?** An instance profile with AmazonSSMManagedInstanceCore, or DHMC enabled and IMDSv2 in use.
- **3. Is there a network path?** Private subnet with no NAT needs **interface endpoints for ssm, ssmmessages and ec2messages**.
- **4. Do the endpoint and subnet allow it?** Security group on the endpoint must permit 443 from the instance.

> ON THE EXAM
>
> Session Manager specifically needs ssmmessages. If Run Command works but sessions fail, that endpoint is the one to check.

Session Manager: what it removes

- **No bastion host.** No SSH keys to distribute, rotate or lose. No RDP.
- **No inbound ports at all.** The agent makes an outbound connection, so security groups need no inbound rules.
- Access is controlled by **IAM**, so joiners and leavers are handled where everything else already is.
- Works on instances in **private subnets with no internet access**, through interface endpoints.

> ON THE EXAM
>
> "Remove the bastion", "stop managing SSH keys" or "no inbound ports" all point at Session Manager. It is one of the most reliable signals on this exam.

Session Manager: the auditable parts

- **Session logs to CloudWatch Logs or S3**, capturing the commands typed and the output returned.
- Every session start and stop is a **CloudTrail event**, tied to the IAM identity.
- **KMS encryption** of session data in transit, above the TLS that is already there.
- **Port forwarding** reaches a database or web console on a private instance, with no tunnel host.

> ON THE EXAM
>
> A bastion tells you someone connected. Session Manager tells you who, when, under which role, and every keystroke.

Session Manager or EC2 Instance Connect?

SESSION MANAGER

Agent-based, no ports

- Needs SSM Agent and a managed node
- No inbound rules, works in private subnets
- Full session logging to S3 or CloudWatch
- Port forwarding and RunAs support

EC2 INSTANCE CONNECT

Pushes a temporary key

- Pushes a one-time SSH public key, valid 60 seconds
- Still an SSH connection on port 22
- Needs an inbound path, or the Endpoint variant
- Linux only, no session logging

THE TIEBREAKER

Both remove long-lived keys. Only Session Manager removes the inbound port and records the session.

Documents: the unit of work

DOCUMENT TYPE	WHAT IT DOES
Command	What Run Command and State Manager execute on a node
Automation	A multi-step workflow over AWS APIs
Session	Configures Session Manager behaviour, such as logging and shell profile
Policy	Used by State Manager, for example to collect Inventory
Package	A Distributor package: software to install on nodes

ON THE EXAM

Documents are versioned and can be shared across accounts, or publicly. AWS-owned ones start with AWS-, such as AWS-RunPatchBaseline.

Run Command: doing something now

- Runs a **document** against nodes chosen by **instance ID, tag, or resource group**. Tags are how you target a fleet.
- **Rate control**: concurrency as a number or percentage, and an **error threshold** that stops the rollout.
- Output goes to **S3 or CloudWatch Logs**. The console truncates at 2,500 characters.
- It is **one-shot**. Nothing re-runs, and instances launched afterwards are untouched.

> ON THE EXAM
>
> The error threshold is the safety feature. Stop after 5 failures and a bad command hits five instances, not five hundred.

Run Command or State Manager?

RUN COMMAND

Imperative, once

- Do this now, on these nodes
- Immediate result, immediate feedback
- Nothing re-applies it
- New instances are not covered

STATE MANAGER

Declarative, continuously

- An association: this is the desired state
- Re-applied on a schedule you choose
- New instances matching the tag are picked up
- Manual changes get corrected

THE TIEBREAKER

"Instances launched later must also be configured", or "the setting keeps drifting back". Both mean State Manager.

Automation runbooks

- A runbook orchestrates **multi-step workflows** across AWS APIs, not just commands inside an instance.
- Hundreds of **AWS-owned runbooks** exist already: restart an instance, create an AMI, resize a volume, remediate a bucket.
- A runbook is a **native EventBridge target**, so an alarm or an API event can trigger one with no Lambda function.
- Supports **approval steps**, rate control across accounts and Regions, and branching on results.

> ON THE EXAM
>
> Run Command acts inside the operating system. Automation acts on AWS resources. That is the cleanest way to separate them.

OpsCenter: one queue for operational work

- An **OpsItem** is a tracked piece of operational work, with status, severity, and the affected resource attached.
- CloudWatch alarms and EventBridge rules can **create OpsItems automatically**, and duplicates are deduplicated.
- Each OpsItem carries **investigation context** and can run an **Automation runbook** to resolve itself.
- Works **across accounts**, so one operations team sees the whole organisation's queue.

> ON THE EXAM
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> This closes the loop from module two. An alarm's Systems Manager action creates an OpsItem here, and the runbook that fixes it is attached to it.

Patch Manager: baselines and approval

- A **patch baseline** defines which patches are approved. AWS provides a default baseline per operating system.
- **Approval rules** auto-approve by classification and severity, usually after a delay such as seven days.
- Explicit **approved and rejected lists** override the rules, for a patch you must have or must not have.
- Operations are **Scan** or **Install**. Scan reports compliance and changes nothing.

> ON THE EXAM
>
> The auto-approval delay is how you let a patch prove itself in the wild for a week before it reaches your fleet.

Patch policies: the recommended method now

- Configured in **Quick Setup**. One wizard covers scanning, installing, baselines and targets.
- Separate **scan and install schedules**: scan daily for compliance, install weekly to limit disruption.
- Applies across **an entire organisation, chosen OUs, or one account and Region**.
- **No patch groups needed.** The older tag-based patch group mechanism is no longer required.

> ON THE EXAM
>
> AWS documents patch policies as the recommended way to configure patching. Maintenance windows and patch groups are the older path.

Inventory, compliance and maintenance windows

- **Inventory** collects installed applications, packages, network config and more, on a schedule, via State Manager.
- **Compliance** reports patch and association status across the fleet, and feeds AWS Config and Security Hub.
- **Maintenance windows** define when disruptive work may run: a schedule, a duration, a cutoff, targets and tasks.
- The **cutoff** stops new tasks starting near the end of the window, so work does not overrun into business hours.

> ON THE EXAM
>
> "Which instances have that vulnerable library?" is Inventory. "Only patch between 2am and 4am on Sunday" is a maintenance window.

Parameter Store

- **Standard tier**: 10,000 parameters, 4 KB each, no charge. **Advanced**: 100,000, 8 KB, charged, plus parameter policies.
- **SecureString** encrypts with KMS. Reading it needs both the parameter permission and the key permission.
- **Hierarchical paths** such as /prod/app/db/host allow IAM policies and GetParametersByPath over a whole branch.
- **Public parameters** supply the latest AMI IDs, so a launch template never needs a hard-coded AMI.

> ON THE EXAM
>
> Parameter policies are Advanced tier only: expire a parameter, or notify before it expires. That is the usual reason to pay.

Parameter Store or Secrets Manager?

PARAMETER STORE WHEN

Configuration

- Endpoints, feature flags, AMI IDs, sizes
- Standard tier is free
- SecureString for the occasional secret
- No built-in rotation

SECRETS MANAGER WHEN

Credentials

- Database passwords, API keys, OAuth tokens
- Automatic rotation, including RDS integration
- Cross-account and cross-Region replication
- Charged per secret per month

THE TIEBREAKER

Does it need to rotate automatically? Yes means Secrets Manager. That single question resolves nearly every instance of this pair.

Beyond EC2: hybrid and multicloud

- **Hybrid activations** register on-premises servers, edge devices and VMs in other clouds as managed nodes.
- An activation has an **expiry** and a **registration limit**. Expiry only blocks new registrations; existing nodes keep working.
- Once registered, **every tool works identically**: Session Manager, Run Command, Patch Manager, Inventory.
- **Fleet Manager** gives a console view of every node: file system, registry, performance counters, RDP and shell.

> ON THE EXAM
>
> Hybrid activations are why "manage on-premises servers the same way" is nearly always a Systems Manager answer.

Four availability changes worth knowing

- **Incident Manager: closed to new customers 7 November 2025.** No new features. AWS recommends **OpsCenter** instead.
- **Change Manager: closed to new customers 7 November 2025.** Existing customers continue.
- **The Systems Manager CloudWatch Dashboard was removed after 30 April 2026.** Use the CloudWatch console.
- **The Advanced Instances Tier was removed on 30 June 2026.** Hybrid and multicloud nodes now register with no per-instance charge.

> ON THE EXAM
>
> From 30 September 2026, Session Manager and Run Command on hybrid nodes move to pay-as-you-go. Registration itself stays free.

A CloudOps engineer launches 20 instances from a current Amazon Linux AMI into a private subnet with no NAT gateway and no internet gateway route. Each instance has an instance profile carrying AmazonSSMManagedInstanceCore.

None of the instances appear in Fleet Manager. The agent is running on all of them. Other instances in a public subnet, with the same profile and AMI, appear normally.

What should the engineer do?

A Add the AmazonSSMFullAccess policy to the instance profile

B Create interface VPC endpoints for ssm, ssmmessages and ec2messages, reachable from the subnet

C Enable Default Host Management Configuration in the account and Region

D Add a NAT gateway so the instances can reach the internet

◆ PAUSE AND DECIDE

What should the engineer do?

A AmazonSSMFullAccess

Permissions are already proven correct: identical instances in a public subnet work with the same profile.

B Interface endpoints for the three services

The only difference between working and failing instances is the network path. Endpoints supply it without exposing the subnet.

C Enable DHMC

Another way to grant permissions, which are not the problem. It also needs the same network path, so nothing changes.

D Add a NAT gateway

Would work, and is the wrong instinct. It gives the whole subnet outbound internet access to solve a three-endpoint problem.

THE SIGNAL

The control group is the clue. Same AMI, same profile, different subnet, different result. Only the network differs.

> A security review requires the company to remove its two bastion hosts. Engineers must still get an interactive shell on Linux and Windows instances in private subnets, every session must be recorded, and access must be governed by existing IAM roles.
>
> The instances are already managed nodes. The team wants the least ongoing administration.

What should the team implement?

A. EC2 Instance Connect, with SSH keys pushed on demand

B. Session Manager, with session logging to CloudWatch Logs

C. A single hardened bastion in a separate account, with CloudTrail enabled

D. AWS Client VPN into the private subnets, with SSH and RDP from laptops

◆ PAUSE AND DECIDE

What should the team implement?

A EC2 Instance Connect

Linux only, so it fails the Windows requirement outright. It also has no session recording and still needs an inbound path.

B Session Manager with logging

Covers Linux and Windows, records every session, uses IAM for access, and needs no inbound ports or hosts to maintain.

C One hardened bastion elsewhere

The review said remove the bastions. Moving one to another account is still a host to patch, and CloudTrail does not record keystrokes.

D Client VPN plus SSH and RDP

Replaces two hosts with a VPN endpoint and per-laptop credentials, and records nothing about what was typed.

THE SIGNAL

Three requirements, and Windows quietly eliminates the closest competitor before recording or IAM are even considered.

A company must scan 900 instances daily for missing patches to keep compliance reporting current, but may only install patches during a two-hour window on Sunday mornings. The fleet spans 12 accounts and 3 Regions in an organisation.

The team currently maintains separate maintenance windows and patch groups in each account, and they frequently drift out of alignment.

What should the team implement?

A. A Quick Setup patch policy for the organisation, with a daily scan schedule and a weekly install schedule

B. A Run Command document invoking AWS-RunPatchBaseline, triggered by an EventBridge schedule in each account

C. One maintenance window per account running AWS-RunPatchBaseline in Install mode daily

D. A State Manager association per account running AWS-RunPatchBaseline in Scan mode hourly

◆ PAUSE AND DECIDE

What should the team implement?

A Organisation-wide patch policy

One configuration across all accounts and Regions, with separate scan and install schedules. It also removes the patch groups that keep drifting.

B Run Command on a schedule per account

Twelve accounts times three Regions of hand-built automation. It rebuilds the drift problem the scenario is complaining about.

C Daily install maintenance windows

Installs daily, which breaks the Sunday-only constraint outright, and keeps the per-account sprawl.

D Hourly scan association

Scans only, so nothing is ever patched. It also scans far more often than the requirement asks for.

THE SIGNAL

Two schedules from one configuration is precisely what patch policies added. The drift complaint is pointing at organisation-level setup.

A team stores an application's database password in Parameter Store as a SecureString and reads it at start-up. A new compliance rule requires the password to change every 30 days without downtime, and the application must always receive the current value.

A second requirement asks that the application's non-sensitive settings, such as the API endpoint and a feature flag, stay where they are at no additional cost.

What should the team do?

A. Move the password to Secrets Manager with automatic rotation, and leave the settings in Parameter Store

B. Upgrade Parameter Store to the Advanced tier and use a parameter policy to expire the password every 30 days

C. Keep everything in Parameter Store and add a Lambda function on an EventBridge schedule to rotate the password

D. Move all parameters, sensitive and not, to Secrets Manager for consistency

◆ PAUSE AND DECIDE

What should the team do?

A Password to Secrets Manager, settings stay

Rotation is the deciding requirement, and RDS integration changes the database and the secret together. Settings stay free in Parameter Store.

B Advanced tier expiry policy

The sharpest distractor. A parameter policy expires or notifies. It deletes the value; it never sets a new password anywhere.

C Lambda rotation function

Rebuilds Secrets Manager rotation by hand, including the risky part where the database and the stored value must change together.

D Everything to Secrets Manager

Charged per secret per month. The second requirement explicitly says the settings must not add cost.

THE SIGNAL

"Expire" and "rotate" are not the same verb. One removes a value; the other replaces it in two places at once.

A CloudWatch alarm on a critical queue's age-of-oldest-message currently notifies an SNS topic. The operations team wants each occurrence tracked as a work item with a status and severity, visible across all 14 accounts. Each item should have a runbook attached that can be run from the item itself.

Repeated firing of the same alarm must not create hundreds of separate items. The team is building this in 2026 and has never used Incident Manager.

What should the team use?

A Systems Manager Incident Manager, with a response plan triggered by the alarm

B Systems Manager OpsCenter, with the alarm configured to create OpsItems

C An EventBridge rule writing alarm events to a DynamoDB table used as a work queue

D Systems Manager Change Manager, with a change template per alarm

◆ PAUSE AND DECIDE

What should the team use?

A Incident Manager response plan

Closed to new customers since 7 November 2025. A team that has never used it cannot enable it, and AWS points them at OpsCenter.

B OpsCenter with OpsItems

Status, severity, cross-account view, attached runbooks and built-in deduplication. Every stated requirement, and it is AWS's recommended path.

C DynamoDB as a work queue

Builds a ticketing system from scratch: no deduplication, no console, no runbook integration, and a table to own for ever.

D Change Manager

Also closed to new customers, and it governs planned changes rather than tracking incidents. Wrong tool and unavailable.

THE SIGNAL

"Building this in 2026 and has never used Incident Manager" is not scene-setting. It is the clause that makes A impossible.

An engineer runs a Run Command document across 800 instances to update a configuration file. A typo in the document corrupts the file, and the instances begin failing health checks. The engineer stops the command after noticing, but 800 instances are already affected.

Management asks what configuration would have limited the damage, and what would keep the file correct from now on.

Which combination answers both questions?

A. Set an error threshold and concurrency limit on Run Command, and use a State Manager association to maintain the file

B. Run the command inside a maintenance window, and enable Inventory collection

C. Use Automation instead of Run Command, and enable Patch Manager compliance reporting

D. Target instances by instance ID rather than by tag, and enable Session Manager logging

◆ PAUSE AND DECIDE

Which combination answers both questions?

A Rate control, plus a State Manager association

Concurrency and an error threshold stop the rollout after a few failures. State Manager then holds the file at its correct content continuously.

B Maintenance window and Inventory

A window controls when it runs, not how many break. Inventory reports what is installed, and would not have caught a corrupted file's contents.

C Automation and patch compliance

Automation acts on AWS resources, not files inside the OS. Patch compliance has nothing to do with a configuration file.

D Target by instance ID

Listing 800 IDs by hand is worse operationally and would still have run on all of them. Session logging records shells, not Run Command.

THE SIGNAL

Rate control is the blast-radius answer; desired state is the drift answer. Two requirements, two different Systems Manager features.

The five things to carry forward

- **Agent, permissions, network.** Every "not a managed node" question is one of those three, and the network one needs three endpoints.
- **DHMC replaces per-instance profiles**, and needs IMDSv2 plus a recent agent.
- **No bastion, no keys, no inbound ports, session recorded.** All four phrases mean Session Manager.
- **Run Command acts inside the OS; Automation acts on AWS resources.** State Manager keeps either one true.
- **Rotation decides Parameter Store versus Secrets Manager.** Expiry is not rotation.

> ON THE EXAM
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> Next: automated remediation and self-healing, which assembles modules 2 to 7 into systems that fix themselves.